

Modified $Prove_{Com}$ of BLNS code

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Reordered internal pseudocode structure

Here I list the line numbers of the Com prove taken from QUBIP pdf, which must be implemented in this order.

N.B. We removed extractable LHC

The lines containing comments are the ones that need to be modified:

```
1. ... 4.
7.
while ( $b_3 == 0$ ) {
    10.
    11.
    12. Sampling  $y_3$  it's sufficient
    15.
    19.  $a_1 = (t_A, t_y)$ 
    20. ... 24.
}
13.
16.
25.  $a_2 = (z_3, t_g)$ 
26. ... 32.
34.
35.
while ( $b_1 b_2 == 0$ ) {
    12. You need to sample  $y_1, y_2$ 
    33.
    14.
    36.
    38.
    37.
    39.  $a_4 = (t, w, f_0)$ 
    40. ... 43.
}
 $a_5 = (z_1, z_2)$ 
 $\pi \leftarrow (a_1, a_2, a_3, a_4, a_5)$ 
 $rst \leftarrow b_1 b_2 b_3$ 
```